

Course Code: CS1103

Name of the Programme: B.Sc. Computer Science (Hons.)

Name of the Course: Operating Systems

Semester: II

Course credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
3	2+0+2	60

Course Objectives:

- a) To equip students with a strong foundational knowledge of OS concepts.
- b) To covers key components of an operating system including, Process management
Memory management, Deadlock detection and handling, Device and I/O management.
- c) To introduce students about Advanced optimization techniques, Scheduling algorithms and mechanisms, System-level resource management.
- d) To emphasize both theoretical understanding and practical application.

	6
Basics of operating systems, Process management Generations, Types, Structure, Services, System Boot, System Programs, Protection and Security. Process Concepts, Process States, Process Control Block, Scheduling-Criteria, Scheduling Algorithms and their Evaluation.	

Module - 2	No. of hours
	6
Thread API Thread Creation, Completion, Locks, Condition Variables, Compilation. Inter Process Communication and Synchronization: Software Approaches, Principles of Concurrency, Hardware Support, Semaphores, Message Passing,	

Module 3	6
System Calls, Traps, Device Drivers, and Deadlocks System Calls – Concepts and Execution Flow, User Mode and Kernel Mode, Trap Handling and Exception Flow, Interrupt Service Routines (ISRs), Implementation of System Calls in xv6, Device Communication and Drivers - Deadlocks – Characterization, Deadlocks in I/O and Device Handling, Prevention, Detection, and Avoidance Techniques in Kernel-Level Resource Management.	

Module - 4	No. of hours
	6
Memory Management Main Memory, Swapping, Contiguous Memory Allocation, Paging, Structure of Page Table, Segmentation, Virtual Memory, Demand Paging, Page Replacement Algorithms, Allocation of Frames, Thrashing.	

Module 5	6
Storage Structure Overview of Mass Storage Structure, Disk Structure, Disk Scheduling, Disk Management, Swap-Space Management, I/O System Overview, I/O Hardware, Application I/O Interface.	

Course outcomes

- a) Understanding of the historical evolution and basic principles of operating systems.

b)	Understand the process management techniques and scheduling.
c)	Analyze deadlock conditions and propose methods to avoid or resolve them.
d)	Analyze and address issues such as memory fragmentation and dynamic memory allocation.

Text Books	
1	Stallings, W. (2018). <i>Operating systems: Internals and design principles</i> (9th ed.). Pearson Education. ISBN 978-9352866717
2	Silberschatz, A., Galvin, P. B., & Gagne, G. (2010). <i>Operating system concepts essentials</i> . John Wiley & Sons. ISBN: 978-0470889206
3	Cox, R., Kaashoek, F., & Morris, R. (2022). <i>xv6: A simple, Unix-like teaching operating system</i> (Rev. 3, RISC-V ed.). MIT CSAIL.

Reference books	
1	Silberschatz, A., Galvin, P. B., & Gagne, G. (2012). <i>Operating system concepts</i> (9th ed.). Wiley. ISBN: 978-1118063330
2	Tanenbaum, A. S., & Bos, H. (2023). <i>Modern operating systems</i> (Global 5th ed.). Pearson. ISBN: 978-1292459660

Lab Programs		[30 Hrs]
1	Process Creation in Unix Shell: Creating Child Processes and Displaying Parent-Child Relationships Using ps.	
2	Process Control: Create a shell script that demonstrates process control commands like bg, fg, jobs, and kill. Implement a simple background task and manage its execution.	
3	Basic Shell Script for Job Scheduling: Develop a shell script that takes a list of commands and executes them with a specified delay using the sleep command. Demonstrate job scheduling in the shell.	
4	File Operations: Create a shell script that performs file operations, such as creating, copying, moving, and deleting files using commands like touch, cp, mv, and rm. Include error handling to manage file operations gracefully.	
5	File Permissions: Write a shell script to modify file permissions and ownership using chmod and chown. Include user input for the filename and desired permissions and display the new file permissions.	
6	Directory Management: Develop a shell script that creates, lists, and deletes directories. Use commands like mkdir, ls, and rmdir, and include user prompts for directory names.	
7	Dynamic Memory Check: Write a shell script that monitors memory usage on the system using commands like free and top. Display the current memory usage and available memory.	
8	Environment Variables: Create a shell script that sets, exports, and displays environment variables. Use printenv and env to show the current environment settings.	
9	I/O Redirection in Unix Shell: Manipulating Data Streams with Enhanced File Operations.	
10	Command Piping in Unix Shell: Exploring Operating System Features.	

